

Testing Hypotheses Using a Mixture of Null and Alternative Models

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September 2026

Abstract

This study examines the testing of statistical hypotheses by forming a mixture of null and alternative models. The motivation for this model expansion stems from the concern that the null hypothesis may be rejected because the original model is misspecified. The likelihood-ratio (LR) test formed by the mixture requires non-standard analysis under the null because the model suffers from twofold identification and boundary parameter problems. We overcome these challenges and show that the null limit distribution of the LR test is represented as a functional of a Gaussian process. We illustrate the use of the LR test by assuming a popular linear model and demonstrate its usefulness by Monte Carlo simulations.

Key Words: Mixture of Null and Alternative Models, Likelihood-Ratio Test, Twofold Identification Problem, Boundary Parameter Problem, Asymptotic Critical Values.

AMS Classification: 62F03, 62F05, 62F30, 62J05.

JEL Classification: C12, C22, C32, C52.

Acknowledgements: Cho acknowledges support from the Chang-am Fellowship.

1 Introduction

Testing statistical hypotheses using standard tests is a popular practice in inferring unknown coefficients. Key coefficient values or their signs often serve to verify the core aspects of maintained hypotheses. For this verification, test statistics obtained by applying Wald, Lagrange multiplier (LM), and likelihood-ratio (LR) test principles are popularly employed for empirical data analysis.

Tests powerful against a wide range of alternatives are desirable for this purpose. Statistical models are often specified by focusing on the maintained hypothesis, containing the parameter value of interest as a special case. Consequently, if the data generating process (DGP) condition is not contained in the model, we should be concerned with the possibility that the null hypothesis may be rejected due to this misspecification.

Maintained hypotheses can take various forms, and there are numerous ways to devise a powerful test for misspecified models. The standard LR test and its asymptotically equivalent analogs, such as Wald and LM tests, are locally most powerful tests if the maintained model assumptions are correct under the regularity conditions provided in [Neyman \(1959\)](#). Otherwise, the power of these tests may be poor. It would be desirable if a standard test were applicable to a model with more alternatives than those aimed at by the original model.

The goal of the current study is therefore to expand the use of the standard LR test by specifying a larger model that treats the original model as a special case. To this end, we form a mixture model of the null and alternative models and test the null hypothesis. For practical use of our approach, we obtain the null limit distribution of the LR test under the mixture model assumption and examine its finite-sample properties by conducting Monte Carlo simulations to compare its size and power with those of the standard LR test.

Mixture models are popular in the literature. In economics, [Heckman and Singer \(1984\)](#) specify mixture of Weibull distributions to estimate unemployment duration data consistently. As another example, [Hamilton \(1989\)](#) develops a mixture of autoregressive processes into a regime switching model and examines the business cycle of the postwar US real GNP. This popularity is not restricted only to economics but also reaches most disciplines, as reviewed by [McLachlan and Basford \(1988\)](#).

Despite its popularity, the prior literature warns against the liberal application of mixture models. As detailed in the next section, forming mixture models with components more than necessary can yield non-standard models. That is, the mixture model may suffer from model identification and interior parameter problems under the null hypothesis that the mixture hypothesis is negated. In particular, the identification problem arises in two different manners. This feature is examined in the prior literature in terms of testing for the mixture model hypothesis via LR test (see [Hartigan, 1985](#); [Ghosh and Sen, 1985](#); [Chernoff and Lander,](#)

1995; Dacunha-Castelle and Gassiat, 1999; Chen and Chen, 2001; Cho and White, 2007, for example), although this hypothesis is different from that of this study.

We exploit the mixture model as a device to expand the original model assumption and provide a methodology to analyze the null limit distribution of the LR test. We suppose that the maintained hypothesis provides a correctly specified model to obtain the limit distribution of the LR test under the null, but also allow for the original model to be possibly misspecified under the alternative so that the mixture model can contain the DGP in the model with a higher chance than the original model. This approach is useful when the empirical researcher is uncertain about how to specify a new model if the null is negated using the original model. In addition, we overcome the challenges of the twofold model identification and boundary parameter problems and represent the null limit distribution of the LR test as a functional of a Gaussian stochastic process defined on the space of the unidentified parameter. Finally, we demonstrate the use of the LR test by using a popular linear model and examine the size and power properties of the LR test by Monte Carlo simulations.

The plan of this paper is as follows. In Section 2, we introduce the regularity conditions for the standard LR test to proceed with our discussions in a manageable way. In Section 3, we expand the original model by forming the mixture of the null and alternative models and test the standard hypothesis using the LR test. We derive the null limit distribution of the LR test in the same section. Monte Carlo simulations are conducted in Section 4, in which we compare the LR test formed by the mixture model with the standard LR test. Section 5 concludes this study, and all mathematical proofs are collected in the Appendix.

Before moving to the next section, we define some mathematical notations. We let " $\rightsquigarrow$ " and " $\xrightarrow{\mathbb{P}}$ " represent "weak convergence" and "convergence in probability", respectively. The matrix $[a_{ij}]$ with $i = 1, \dots, m, j = 1, \dots, n$ is an $m \times n$ matrix with the element a_{ij} , and we let $\|\mathbf{A}\|$ be the Frobenius norm of $\mathbf{A}$, viz., $\|\mathbf{A}\| = [\text{tr}(\mathbf{A}\mathbf{A}')]^{1/2}$. If $\mathbf{a}$ is a vector, $\|\mathbf{a}\|$ is the same as the Euclidean norm of $\mathbf{a}$. We also let $\|\mathbf{A}\|_\infty$ denote the uniform norm of $\mathbf{A}$. All others are standard.

2 DGP and Models

In this section, we motivate the current study. For this purpose, we impose a set of regularity conditions. We suppose the following DGP as the first condition:

Assumption 1. (i) $\{\mathbf{x}_t \in \mathbb{R}^d\}_{t=1}^n$ ($d \in \mathbb{N}$) is a sequence of strictly stationary β -mixing processes on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that for some $c > 0$ and $\rho \in [0, 1)$, the β -mixing coefficient β_τ is at most $c\rho^\tau$;

(ii) for each $t = 1, \dots, n$, the conditional distribution function of $\mathbf{x}_t$ on $\mathbf{x}^{t-1}$ denoted as $g(\cdot|\mathbf{x}^{t-1})$ is a measurable function almost surely (a.s.) $-\mathbb{P}$, where $\mathbf{x}^{t-1} := (\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \dots)$. $\square$

The β -mixing condition is crucial in our context. As detailed below, we utilize this DGP condition to obtain the weak limit of the LR test under the null. Specifically, the LR test defined in the current study converges weakly to a functional of a Gaussian stochastic process under the null, and the β -mixing condition is crucial for this purpose.

We next impose the following model condition:

Assumption 2. (i) $\mathcal{M}_o := \{f(\cdot|\mathbf{x}^{t-1}; \boldsymbol{\theta}, \boldsymbol{\xi}) : (\boldsymbol{\theta}, \boldsymbol{\xi}) \in \boldsymbol{\Theta} \times \boldsymbol{\Xi}\}$ is specified as a model for $g(\cdot|\mathbf{x}^{t-1})$, where $\boldsymbol{\Theta}$ and $\boldsymbol{\Xi}$ are convex and compact sets in $\mathbb{R}^{r_1}$ and $\mathbb{R}^{r_2}$ respectively ($r_1, r_2 \in \mathbb{N}$);
(ii) $f_t(\cdot) \in \mathcal{C}^{(2)}(\boldsymbol{\Theta} \times \boldsymbol{\Xi})$ a.s. $-\mathbb{P}$, where $\mathcal{C}^{(2)}(\cdot)$ is a set of twice continuously differentiable functions defined on the given argument, where $f_t(\cdot)$ denotes $f(\mathbf{x}_t|\mathbf{x}^{t-1}; \cdot)$. $\square$

The model condition in Assumption 2 is a parametric specification, in which the parameter is separated into two groups. The first parameter $\boldsymbol{\theta}$ is a set of parameters of interest to be tested by a test statistic, and $\boldsymbol{\xi}$ is a nuisance parameter. For notational simplicity, we let $\boldsymbol{\gamma}$ and $\boldsymbol{\Gamma}$ denote $(\boldsymbol{\theta}, \boldsymbol{\xi})$ and $\boldsymbol{\Theta} \times \boldsymbol{\Xi}$ respectively.

We further assume that the given model satisfies the following additional regularity conditions:

Assumption 3. (i) For each $\boldsymbol{\gamma} := (\boldsymbol{\theta}, \boldsymbol{\xi}) \in \boldsymbol{\Gamma} := \boldsymbol{\Theta} \times \boldsymbol{\Xi}$, $\mathbb{E}[\ell_t^o(\boldsymbol{\gamma})]$ exists, and $\mathbb{E}[\ell_t^o(\cdot)]$ is finite uniformly on $\boldsymbol{\Gamma}$, where for each $\boldsymbol{\gamma}$, $\ell_t^o(\boldsymbol{\gamma}) := \log[f_t(\boldsymbol{\gamma})]$;
(ii) $\boldsymbol{\gamma}_* := (\boldsymbol{\theta}_*, \boldsymbol{\xi}_*)$ uniquely maximizes $\mathbb{E}[\ell_t^o(\cdot)]$ on $\boldsymbol{\Gamma}$ and is interior to $\boldsymbol{\Gamma}$. $\square$

Assumption 4. For each $i_1, i_2 \in \{\theta_1, \dots, \theta_{r_1}, \xi_1, \dots, \xi_{r_2}\}$, there exists a sequence of positive, strictly stationary, and ergodic processes $\{M_t\}$ such that for some $\delta > 0$, $\mathbb{E}[M_t^{1+\delta}] < \Delta < \infty$, $\sup_{\boldsymbol{\gamma}} |\nabla_{i_1} \ell_t^o(\boldsymbol{\gamma})|^2 \leq M_t$, and $\sup_{\boldsymbol{\gamma}} |\nabla_{i_1} \nabla_{i_2} \ell_t^o(\boldsymbol{\gamma})| \leq M_t$. $\square$

Assumption 5. The asymptotic covariance matrix of $\{n^{-1/2} \nabla_{\boldsymbol{\gamma}} L_n^o(\boldsymbol{\gamma}_*)\}$ is positive definite, and $n^{-1} \mathbb{E}[\nabla_{\boldsymbol{\gamma}}^2 L_n^o(\boldsymbol{\gamma}_*)]$ is negative definite whenever they exist, where for each $\boldsymbol{\gamma} \in \boldsymbol{\Gamma}$, $L_n^o(\boldsymbol{\gamma}) := \sum_{t=1}^n \ell_t^o(\boldsymbol{\gamma})$. $\square$

For notational simplicity, we let $\mathbf{A}_* := n^{-1} \mathbb{E}[\nabla_{\boldsymbol{\gamma}}^2 L_n^o(\boldsymbol{\gamma}_*)]$ and $\mathbf{B}_* := \text{avar}[n^{-1/2} \nabla_{\boldsymbol{\gamma}} L_n^o(\boldsymbol{\gamma}_*)]$, where ‘avar’ stands for ‘asymptotic covariance’.

These regularity conditions are standard in the prior literature. The unknown parameters are estimated by the extremum estimator defined as

$$\hat{\boldsymbol{\gamma}}_n^o := (\hat{\boldsymbol{\theta}}_n^o, \hat{\boldsymbol{\xi}}_n^o) := \arg \max_{(\boldsymbol{\theta}, \boldsymbol{\xi}) \in \boldsymbol{\Theta} \times \boldsymbol{\Xi}} L_n^o(\boldsymbol{\theta}, \boldsymbol{\xi}),$$

and its limit distribution is obtained using the regularity conditions given above. Here, the superscript ‘ o ’ denotes the parameter estimator obtained from the current original model. Furthermore, when testing the following hypotheses: for some interior element θ_0 in Θ ,

$$\mathcal{H}_0 : \theta_* = \theta_0; \text{ versus } \mathcal{H}_1 : \theta_* \neq \theta_0,$$

the null limit distribution of a test, say LR test, is determined by the above regularity conditions.

We specifically detail the limit distribution of the extremum estimator and the null limit distribution of the LR test. For this, if we let $\tilde{\gamma}_n^o := (\theta_0, \tilde{\xi}_n^o) := \arg \max_{\xi \in \Xi} L_n^o(\theta_0, \xi)$, then the standard LR test is defined as

$$\mathcal{LR}_n := 2\{L_n(\hat{\gamma}_n^o) - L_n^o(\tilde{\gamma}_n^o)\}.$$

The limit distribution of the extremum estimator $\hat{\gamma}_n$ and the null limit distribution of the LR test are contained in the following theorem:

Theorem 1. *Given Assumptions 1, 2, 3, 4, and 5,*

- (i) $\sqrt{n}(\hat{\gamma}_n - \gamma_*) \overset{A}{\rightsquigarrow} \mathcal{N}(0, C_*)$, where $C_* := A_*^{-1} B_* A_*^{-1}$;
- (ii) if $\mathcal{M}_o$ is correctly specified, $\mathcal{LR}_n \rightsquigarrow z'z$ under $\mathcal{H}_0$, where $z \sim \mathcal{N}(0, I_{r_1})$. □

These properties essentially follow by applying a quadratic Taylor’s expansion to the log-likelihood function. As they are well known, we do not prove Theorem 1 here (e.g., [White, 2001](#)). It is also well known that the LR test is asymptotically and locally optimal test.

Despite the regular asymptotic behaviors of the estimator and test, the maintained model assumption may be too narrow to specify a correct model. This concern arises from typical statistical practice. The empirical researcher is often concerned with the possibility that the null may be rejected because the model $\mathcal{M}_o$ may be misspecified.

We resolve this concern by expanding the model scope to a larger model so that the original model $\mathcal{M}_o$ nests in the larger model. There can be a number of ways to achieve this, but we here focus on a mixture model specified as follows:

$$\mathcal{M} := \{\pi f_t(\theta, \xi) + (1 - \pi)f_t(\theta_0, \xi) : (\pi, \theta, \xi) \in [0, 1] \times \Theta \times \Xi\}.$$

This is a mixture of the null and alternative models, and it is different from the standard mixture in the sense that the parameter of interest, viz., θ , is fixed at the null parameter value θ_0 . The standard mixture model does not fix the parameters and instead estimates them by the maximum-likelihood (ML) estimation.

In the prior literature, mixture models similar to our mixture model are used for other purposes. [Hartigan](#)

(1985) illustrates a mixture of normals with known and unknown parameters to emphasize the role of a compact parameter space when the LR test is used for the mixture hypothesis. Liu and Shao (2003, example 3.1) also consider a similar mixture model to illustrate their theory on general quadratic approximation. Our use of the mixture model is different from theirs. We employ the mixture model to test $\mathcal{H}_0$ against $\mathcal{H}_1$.

There exist other motivations to exploit the mixture model to expand the model scope. First, observational outliers are often believed to lead to distributional misspecification, which may prevent consistent estimation of the parameter of interest (see Heckman and Singer, 1984, for example). We aim to detect the distribution of outliers by specifying the mixture. Second, unobserved heterogeneity also modifies the distributional aspect of observed data. In the literature of unobserved heterogeneity, mixtures of two distributions are popularly specified (see Van den Berg and Ridder, 1998; Cho and Han, 2009; Cho and White, 2010; Cho, 2025, 2026, among others). From this we aim to complement the power loss of a test by using the mixture model. Mixture models are flexible distributional specifications designed to remedy distributional misspecification.

3 LR Test

In this section, we develop a testing procedure based on the LR test principle using the mixture model. To this end, we focus on deriving the null limit distribution of the LR test.

Before proceeding with our further discussion, we first provide further regularity conditions under which we can derive the null limit distribution of the LR test. First, we let for each $(\pi, \gamma) \in [0, 1] \times \mathbf{\Gamma}$,

$$\ell_t(\pi, \gamma) := \log[\pi f_t(\boldsymbol{\theta}, \boldsymbol{\xi}) + (1 - \pi)f_t(\boldsymbol{\theta}_0, \boldsymbol{\xi})] \quad (1)$$

and impose the following additional conditions:

Assumption 6. (i) For every (π, γ) , $\mathbb{E}[\ell_t(\pi, \gamma)]$ exists and is finite uniformly on $[0, 1] \times \mathbf{\Gamma}$;

(ii) $(\pi_*, \gamma_*) := (\pi_*, \boldsymbol{\theta}_*, \boldsymbol{\xi}_*)$ maximizes $\mathbb{E}[\ell_t(\pi, \gamma)]$ on $[0, 1] \times \mathbf{\Gamma}$. □

Assumption 7. There exists a sequence of positive, strictly stationary, and ergodic random variables $\{M_t\}$ such that for some $\delta > 0$, $\mathbb{E}[M_t^{1+\delta}] < \Delta < \infty$, and for each $i_1, i_2 \in \{\pi, \theta_1, \dots, \theta_{r_1}, \xi_1, \dots, \xi_{r_2}\}$, $\sup_{(\pi, \gamma)} |\nabla_{i_1} \ell_t(\pi, \gamma)|^2 \leq M_t$ and $\sup_{(\pi, \gamma)} |\nabla_{i_1} \nabla_{i_2} \ell_t(\pi, \gamma)| \leq M_t$. □

Assumptions 6 and 7 play roles analogous to those of Assumptions 3 and 4, respectively, in the context of the mixture model. Using them, we can show the asymptotic behavior of the following extremum estimator:

$$(\hat{\pi}_n, \hat{\gamma}_n) := (\hat{\pi}_n, \hat{\boldsymbol{\theta}}_n, \hat{\boldsymbol{\xi}}_n) := \arg \max_{(\pi, \boldsymbol{\theta}, \boldsymbol{\xi}) \in [0, 1] \times \boldsymbol{\Theta} \times \boldsymbol{\Xi}} L_n(\pi, \boldsymbol{\theta}, \boldsymbol{\xi}),$$

where $L_n(\pi, \theta, \xi) := \sum_{t=1}^n \ell_t(\pi, \theta, \xi)$.

Theorem 2. *Given Assumptions 1, 2, 6, and 7, $(\hat{\pi}_n, \hat{\gamma}_n) \xrightarrow{\mathbb{P}} (\pi_*, \gamma_*)$.* □

The convergence limit in Theorem 2 needs to be carefully treated. It does not necessarily imply that the convergence limit of the extremum estimator is unique. Assumption 6 does not require that (π_*, γ_*) is unique in $[0, 1] \times \Gamma$. That is, $\mathbb{E}[\ell_t(\cdot)]$ can be equally maximized by multiple parameters, so that the model $\mathcal{M}$ can be unidentified under Assumption 6. We denote this using the following set:

$$\Upsilon := \{(\pi, \gamma) \in [0, 1] \times \Gamma : \mathbb{E}[\ell_t(\pi, \gamma)] = \mathbb{E}[\ell_t(\pi_*, \gamma_*)]\},$$

which collects the parameters attaining the maximum level of $\mathbb{E}[\ell_t(\cdot, \cdot)]$. Assumption 6 allows the number of elements in Υ to exceed a singleton. Accordingly, Theorem 2 should be interpreted such that the minimum distance between the extreme estimator and Υ converges to zero in probability, and (π_*, γ_*) in Theorem 2 is a representative element of Υ .

Indeed, we cannot assume that Υ has a singleton given our interest. We first note that the original null parameter value θ_0 can be related to Υ . If the model is correctly specified under the null, then $\mathbb{E}[\ell_t(\theta_0, \xi_*)] = \mathbb{E}[\ell_t(\pi_*, \gamma_*)]$, so that it follows that

$$\Upsilon = \{(\pi, \theta, \xi_*) \in [0, 1] \times \Theta \times \Xi : \theta = \theta_0 \text{ or } \pi = 0\}.$$

From this, the original null parameter value is associated with Υ in the context of the mixture model. We further partition the null hypothesis into two exclusive subsets as follows:

$$\mathcal{H}_{01} : \theta_* = \theta_0 \text{ and } \pi_* \neq 0 \quad \text{and} \quad \mathcal{H}_{02} : \theta_* \neq \theta_0 \text{ and } \pi_* = 0.$$

The standard regularity condition requiring model identification cannot coexist with the given null hypothesis. Under $\mathcal{H}_{01}$, if $\theta_* = \theta_0$, then π does not matter in maximizing $n^{-1}\mathbb{E}[L_n(\cdot, \cdot)]$, so that π_* is not identified. This implies that Davies's (1977; 1987) identification problem arises. Similarly, another identification problem arises under $\mathcal{H}_{02}$. That is, θ does not matter in maximizing $n^{-1}\mathbb{E}[L_n(\cdot, \cdot)]$ under $\mathcal{H}_{02}$. We call this twofold identification problem. Furthermore, π_* is on the boundary of the parameter space $[0, 1]$ under $\mathcal{H}_{02}$, which also violates the standard interiority condition. Because of these problems, the null limit distribution of the LR test is different from the standard chi-squared distribution.

A few remarks are warranted for the alternative model $\mathcal{M}$. First, the mixture model in $\mathcal{M}$ covers a wider range of alternatives than $\mathcal{M}_o$. This can be verified by letting $\pi = 1$. That is, the original model $\mathcal{M}_o$ is a subset of this mixture, and more alternatives are included in $\mathcal{M}$. Second, $\mathcal{M}$ is specified by using the null

parameter value θ_0 . Unless it is specified as a known parameter, testing the original hypothesis reduces to testing for a mixture hypothesis, leading to an inference problem different from testing $\mathcal{H}_0$. We avoid this issue by fixing one of the parameters at its null value θ_0 .

We examine below the limit distributions of the LR statistic under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$ and combine them to derive the limit distribution of the LR test under $\mathcal{H}_0$.

3.1 LR Test under $\mathcal{H}_{01}$

The standard LR test needs a careful treatment due to the identification problem mentioned above so that it can be analyzed by approximating the log-likelihood function by a quadratic function under $\mathcal{H}_{01}$. We first note that π_* is not identified under $\mathcal{H}_{01}$, so that we first maximize $L_n(\cdot, \cdot)$ with respect to γ and then with respect to π to analyze the LR test appropriately. That is, for each $\pi \in (0, 1]$, we let

$$\hat{\gamma}_n(\pi) := \arg \max_{\gamma \in \Gamma} L_n(\pi, \gamma),$$

and note that its corresponding LR test can be defined as

$$\mathcal{LR}'_n := \max_{\pi \in (0, 1]} 2 \{L_n(\pi, \hat{\gamma}_n(\pi)) - L_n^o(\tilde{\gamma}_n^o)\}.$$

We here note that $\hat{\gamma}_n(\pi)$ is an estimator obtained by fixing π . If π is fixed, the identification problem does not arise, so that $L_n(\pi, \cdot)$ can be approximated by a quadratic function. Specifically, for each π , we obtain the following approximation:

$$2 \{L_n(\pi, \gamma) - L_n(\pi, \gamma_*)\} = \frac{2}{\sqrt{n}} \nabla'_\gamma L_n(\pi, \gamma_*) \delta_n + \delta'_n \left\{ \frac{1}{n} \nabla_\gamma^2 L_n(\pi, \gamma_*) \right\} \delta_n + o_{\mathbb{P}\pi}(1), \quad (2)$$

where $\delta_n := \sqrt{n}(\gamma - \gamma_*)$.

The given approximation is related to the score and Hessian matrix obtained from the original model $\mathcal{M}_o$. First, the first-order derivative $\nabla_\gamma L_n(\pi, \gamma_*)$ is related to the score obtained from $\mathcal{M}_o$. For each π , we note that $\nabla_\gamma L_n(\pi, \gamma_*) = \mathbf{I}(\pi) \nabla_\gamma L_n^o(\gamma_0)$ under $\mathcal{H}_{01}$ by noting $\xi_* = \xi_*$ under $\mathcal{H}_{01}$, where $\gamma_0 := (\theta_0, \xi_*)$ and

$$\mathbf{I}(\pi) := \begin{bmatrix} \pi \mathbf{I}_{r_1} & 0 \\ 0 & \mathbf{I}_{r_2} \end{bmatrix}.$$

Second, the second-order derivative $\nabla_\gamma^2 L_n(\pi, \gamma_*)$ is related to the Hessian matrix obtained from $\mathcal{M}_o$. For each π , it follows that

$$\nabla_\gamma^2 L_n(\pi, \gamma_*) = \mathbf{I}(\pi) \nabla_\gamma^2 L_n^o(\gamma_0) \mathbf{I}(\pi) + o_{\mathbb{P}\pi}(n)$$

Under $\mathcal{H}_{01}$. Therefore, $n^{-1}\nabla_{\gamma}^2 L_n(\pi, \gamma_{\star}) \xrightarrow{\mathbb{P}} \mathbf{I}(\pi) \mathbf{A}_{\star}^{-1} \mathbf{I}(\pi)$ by the ergodic theorem. The remainder term $o_{\mathbb{P}\pi}(n)$ depends on π , but it is $o_{\mathbb{P}}(n)$ uniformly in π because the nuisance parameter exists as a linear coefficient and the second-order derivative is uniformly bounded by M_t in Assumption 7.

It also turns out that even the LR test $\mathcal{LR}'_n$ is related to the original LR test $\mathcal{LR}_n$ under $\mathcal{H}_{01}$. For each π , we rewrite (2) as

$$2 \{L_n(\pi, \gamma) - L_n(\pi, \gamma_0)\} = \frac{2}{\sqrt{n}} \nabla'_{\gamma} L_n^o(\gamma_0) \mathbf{I}(\pi) \delta_n + \delta'_n \left\{ \frac{1}{n} \mathbf{I}(\pi) \nabla_{\gamma}^2 L_n^o(\gamma_0) \mathbf{I}(\pi) \right\} \delta_n + o_{\mathbb{P}}(1)$$

under $\mathcal{H}_{01}$, where the remainder is $o_{\mathbb{P}}(1)$ uniformly in π because the given model is twice continuously differentiable and the second derivative is uniformly bounded by M_t in Assumption 7. Maximizing the right side with respect to δ_n gives rise to the estimator asymptotically equivalent to $\sqrt{n}(\hat{\gamma}_n(\pi) - \gamma_0)$ and

$$\begin{aligned} 2 \{L_n(\pi, \hat{\gamma}_n(\pi)) - L_n(\pi, \gamma_0)\} &= \{\nabla'_{\gamma} L_n^o(\gamma_0)\} \{-\nabla_{\gamma}^2 L_n^o(\gamma_0)\}^{-1} \{\nabla_{\gamma} L_n^o(\gamma_0)\} + o_{\mathbb{P}}(1) \\ &= \sup_{\gamma \in \Gamma} 2 \{L_n^o(\gamma) - L_n^o(\gamma_0)\} + o_{\mathbb{P}}(1). \end{aligned}$$

The right side is independent of the nuisance parameter π . That is, maximizing it further with respect to π is an innocuous process, yielding the following:

$$\sup_{\pi \in (0,1]} 2 \{L_n(\pi, \hat{\gamma}_n(\pi)) - L_n(\pi, \gamma_0)\} = \{\nabla'_{\gamma} L_n^o(\gamma_0)\} \{-\nabla_{\gamma}^2 L_n^o(\gamma_0)\}^{-1} \{\nabla_{\gamma} L_n^o(\gamma_0)\} + o_{\mathbb{P}}(1). \quad (3)$$

Given this, we further note the following to relate $\mathcal{LR}'_n$ to (3):

$$\mathcal{LR}'_n = \sup_{\pi \in (0,1]} 2 \{L_n(\pi, \hat{\gamma}_n(\pi)) - L_n(\pi, \gamma_0)\} - 2 \{L_n^o(\hat{\gamma}_n^o) - L_n^o(\gamma_0)\}$$

because for every π , $L_n^o(\gamma_0) = L_n(\pi, \gamma_0)$. The following theorem then follows:

Theorem 3. *Given Assumptions 1, 2, 5, 6, and 7, if $\mathcal{M}_o$ is correctly specified,*

- (i) $\mathcal{LR}'_n \rightsquigarrow \mathbf{z}' \mathbf{z}$; and
- (ii) $\mathcal{LR}'_n - \mathcal{LR}_n \xrightarrow{\mathbb{P}} 0$ under $\mathcal{H}_{01}$. □

Remark 1. (i) As Theorem 3 (ii) states, $\mathcal{LR}'_n$ is asymptotically equivalent to $\mathcal{LR}_n$ in Theorem 1. This aspect implies that testing the original hypotheses using the mixture (1) does not signal the effect of model expansion under $\mathcal{H}_{01}$. A different null limit distribution would have been obtained for $\mathcal{LR}'_n$ if the model expansion were effective.

- (ii) In the pure mixture context, when testing for the mixture hypothesis, the log-likelihood function cannot be approximated by a second-order approximation as given in (2). A fourth- or eight-order

approximation is required, because the score or Hessian matrix are identical to zero. In the current study, the second-order approximation suffices for model approximation mainly because θ_0 is fixed instead of being estimated, ensuring that the score differs from zero with probability 1. From this, the convergence rate of the extremum estimator becomes $\sqrt{n}$. $\square$

3.2 LR Test under $\mathcal{H}_{02}$

In this section, we examine the LR test under $\mathcal{H}_{02}$. As it turns out, the LR test under $\mathcal{H}_{02}$ essentially achieves the goal of this paper by demonstrating that its null limit distribution differs from that of $\mathcal{LR}_n$.

The prior regularity condition, particularly Assumption 5 needs to be elaborated on for the goal of this section. Under $\mathcal{H}_{02}$, Assumption 5 needs to be strengthened to obtain the null limit distribution of the LR test because θ_* is not identified. Under $\mathcal{H}_{01}$, π_* is not identified and it exists as a linear coefficient, but θ_* does not exist as a linear coefficient, so that it is not canceled out when optimizing the likelihood function with respect to the unidentified parameter. This also makes it necessary to regularize the log-likelihood function to handle the nonlinearity driven by θ_* . For this purpose, we let $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ be the minimum and maximum eigenvalues of the given matrix, respectively and assume the following:

Assumption 8. (i) $\lambda_{\max}(n^{-1}\mathbb{E}[\nabla_{(\pi,\gamma)}^2 L_n(\pi_*, \gamma_*)]) \leq 0$ such that $\lambda_{\min}(n^{-1}\mathbb{E}[\nabla_{(\pi,\gamma)}^2 L_n(\pi_*, \gamma_*)]) = 0$ if and only if $\pi_* = 0$ or $\theta_* = \theta_0$;
(ii) for each $(\pi_*, \gamma_*) \in \{(\pi, \gamma) \in \Upsilon : \pi_* = 0, \theta_* \neq \theta_0\}$,

$$\lambda_{\max}(n^{-1}\mathbb{E}[\nabla_{\omega}^2 L_n(\pi_*, \gamma_*)]) < 0 \quad \text{and} \quad \lambda_{\min}(\text{avar}[n^{-1/2}\nabla_{\omega} L_n(\pi_*, \gamma_*)]) > 0,$$

where $\omega := (\pi, \xi)$;

(iii) for each $(\pi_*, \gamma_*) \in \{(\pi, \gamma) \in \Upsilon : \pi_* \neq 0, \theta_* = \theta_0\}$,

$$\lambda_{\max}(n^{-1}\mathbb{E}[\nabla_{\gamma}^2 L_n(\pi_*, \gamma_*)]) < 0 \quad \text{and} \quad \lambda_{\min}(\text{avar}[n^{-1/2}\nabla_{\gamma} L_n(\pi_*, \gamma_*)]) > 0. \quad \square$$

Assumption 5 is implied by Assumption 8. In addition, θ_* is not identified under $\mathcal{H}_{02}$ as we discussed above. For our purpose, we restrict the model in a way it is identified if $\theta \neq \theta_*$ by assuming that $\lambda_{\max}(n^{-1}\mathbb{E}[\nabla_{\omega}^2 L_n(\theta, \omega_*)]) < 0$ under $\mathcal{H}_{02}$. This condition is the minimum level restriction to analyze the unidentified model under the null.

As the LR test suffers from an identification problem under $\mathcal{H}_{02}$, optimizing $L_n(\cdot, \cdot)$ needs a careful

treatment. We first optimize the likelihood function with respect to ω , so that for each θ , we let

$$\hat{\omega}_n(\theta) := \left(\hat{\pi}_n(\theta), \hat{\xi}_n(\theta) \right) := \arg \max_{\omega \in [0,1] \times \Xi} \bar{L}_n(\theta, \omega),$$

where for every (π, γ) , $\bar{L}_n(\theta, \omega) := L_n(\pi, \gamma)$. As for $\hat{\gamma}_n(\pi)$, $\hat{\omega}_n(\theta)$ is an estimator obtained by fixing θ . Using this, we now define the following LR test:

$$\mathcal{LR}_n''(\epsilon) := \sup_{\theta \in \Theta(\epsilon)} 2 \left\{ \bar{L}_n(\theta, \hat{\omega}_n(\theta)) - \bar{L}_n(\theta, \tilde{\omega}_n^o) \right\},$$

where $\Theta(\epsilon) := \{\theta \in \Theta : \|\theta - \theta_0\| \geq \epsilon\}$. This LR test corresponds to $\mathcal{LR}_n'$, and we exclude θ_0 from Θ so that the LR test under $\mathcal{H}_{02}$ can be distinguished from $\mathcal{LR}_n'$, which we call the ϵ -restriction. If θ_0 is included in the parameter space, it is not straightforward how the LR tests under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$ differ from each other as $\mathcal{H}_{01}$ can be treated as a special case of $\mathcal{H}_{02}$ by fixing the unidentified parameter at θ_0 . We remove θ_0 from our consideration for the moment and examine the separate limit behaviors of the LR tests under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$. After this, we below modify $\Theta(\epsilon)$ to Θ and examine the null limit distribution of the LR test.

We now analyze the LR test under $\mathcal{H}_{02}$ by approximating the log-likelihood function by a quadratic function. First, θ_* is not identified under $\mathcal{H}_{02}$, so that for each θ ,

$$2\{\bar{L}_n(\theta, \omega) - \bar{L}_n(\theta, \omega_*)\} = \frac{2}{\sqrt{n}} \nabla'_{\omega} \bar{L}_n(\theta, \omega_*) \bar{\delta}_n + \bar{\delta}_n' \left\{ \frac{1}{n} \nabla_{\omega}^2 \bar{L}_n(\theta, \omega_*) \right\} \bar{\delta}_n + o_{\mathbb{P}}(1),$$

where $\bar{\delta}_n := \sqrt{n}(\omega - \omega_*)$. Here, we note that $\xi_* = \xi_*$ under $\mathcal{H}_{02}$, implying that the second-order Taylor expansion is given as

$$2\{\bar{L}_n(\theta, \omega) - \bar{L}_n(\theta, \omega_0)\} = \frac{2}{\sqrt{n}} \nabla'_{\omega} \bar{L}_n(\theta, \omega_0) \bar{\delta}_n + \bar{\delta}_n' \left\{ \frac{1}{n} \nabla_{\omega}^2 \bar{L}_n(\theta, \omega_0) \right\} \bar{\delta}_n + o_{\mathbb{P}}(1),$$

where $\omega_0 = (0, \xi_*)$, and the remainder term is $o_{\mathbb{P}}(1)$ uniformly in θ by the differentiability condition in Assumption 2 and the boundedness condition in Assumption 7. Therefore, maximizing the left side with respect to ω is asymptotically equivalent to maximizing the right side with respect to $\bar{\delta}_n$. That is, for each θ , we obtain the following approximation of the LR test:

$$\begin{aligned} 2\{\bar{L}_n(\theta, \hat{\omega}_n(\theta)) - \bar{L}_n(\theta, \omega_0)\} &= \max \left[0, \left\{ -\nabla_{\pi}^2 \bar{L}_n(\theta, \omega_0) \right\}^{-1/2} \nabla_{\pi} \bar{L}_n(\theta, \omega_0) \right]^2 \\ &\quad - \nabla'_{\xi} \bar{L}_n(\theta, \omega_0) \left\{ \nabla_{\xi}^2 \bar{L}_n(\theta, \omega_0) \right\}^{-1} \nabla_{\xi} \bar{L}_n(\theta, \omega_0) + o_{\mathbb{P}}(1), \end{aligned} \quad (4)$$

where for each (θ, ω) , we let

$$\nabla_{\pi} \bar{L}_n(\theta, \omega) := \nabla_{\pi} \bar{L}_n(\theta, \omega) - \nabla_{\pi} \nabla'_{\xi} \bar{L}_n(\theta, \omega) \left\{ \nabla_{\xi}^2 \bar{L}_n(\theta, \omega) \right\}^{-1} \nabla_{\xi} \bar{L}_n(\theta, \omega) \quad \text{and}$$

$$\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}) := \nabla_{\pi}^2 \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}) - \nabla_{\pi} \nabla_{\xi}' \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}) \{ \nabla_{\xi}^2 \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}) \}^{-1} \nabla_{\xi} \nabla_{\pi} \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}).$$

This maximization procedure is different from the standard case because ‘max’ operator is involved with it. This usage is motivated by the fact that $\bar{\boldsymbol{\delta}}_n$ has to be constrained to $\bar{\boldsymbol{\Delta}}_n := \{ \sqrt{n}(\pi, \boldsymbol{\xi} - \boldsymbol{\xi}_*) : (\pi, \boldsymbol{\xi}) \in [0, 1] \times \Xi \}$, where $\sqrt{n}\pi$ has to be always greater than zero. This boundary parameter problem results in the use of the ‘max’ operator.

Second, the asymptotic null approximation in (4) is related to the LR statistic testing $\boldsymbol{\omega}$. We note that for each $\boldsymbol{\theta}$, the LR test for testing $\mathcal{H}_{02}$ is constructed as

$$\begin{aligned} & 2 \{ \bar{L}_n(\boldsymbol{\theta}, \hat{\boldsymbol{\omega}}_n(\boldsymbol{\theta})) - \bar{L}_n(\boldsymbol{\theta}, \tilde{\boldsymbol{\omega}}_n^o) \} \\ &= 2 \{ \bar{L}_n(\boldsymbol{\theta}, \hat{\boldsymbol{\omega}}_n(\boldsymbol{\theta})) - \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \} - 2 \{ \bar{L}_n(\boldsymbol{\theta}, \tilde{\boldsymbol{\omega}}_n^o) - \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \} \\ &= \max \left[0, \left\{ -\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \right\}^{-1/2} \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \right]^2 + o_{\mathbb{P}}(1) \end{aligned} \quad (5)$$

under $\mathcal{H}_{02}$, where $\tilde{\boldsymbol{\omega}}_n^o := (0, \tilde{\boldsymbol{\xi}}_n^{o'})'$, because

$$2 \{ \bar{L}_n(\boldsymbol{\theta}, \tilde{\boldsymbol{\omega}}_n^o) - \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \} = \nabla_{\xi}' \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \{ -\nabla_{\xi}^2 \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \}^{-1} \nabla_{\xi} \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) + o_{\mathbb{P}}(1)$$

by a second-order Taylor’s approximation. Here, we note that this LR statistic tests the hypothesis that $\boldsymbol{\omega}_* = \boldsymbol{\omega}_0$ and also asymptotically cancels the second term on the right side of (4), from which (5) follows.

Given this derivation, we obtain the null limit distribution of the LR test by applying the functional central limit theorem (FCLT) and uniform law of large numbers (ULLN). We note that $\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)$ and $\nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)$ are the realized levels of $\nabla_{\pi}^2 \ddot{L}_n(\cdot, \boldsymbol{\omega}_0)$ and $\nabla_{\pi} \ddot{L}_n(\cdot, \boldsymbol{\omega}_0)$, respectively, when $\boldsymbol{\theta}$ is given, which obey the ULLN and FCLT, respectively as follows:

Lemma 1. *Given Assumptions 1, 2, 6, 7, 8, and $\mathcal{H}_{02}$, if $\mathcal{M}_o$ is correctly specified, then for each $\epsilon > 0$,*

- (i) $n^{-1/2} \nabla_{\omega} \bar{L}_n(\cdot, \boldsymbol{\omega}_0) \rightsquigarrow \bar{\mathcal{G}}(\cdot)$, where $\bar{\mathcal{G}} : \Theta(\epsilon) \mapsto \mathbb{R}^{1+r_2}$ is a Gaussian process such that for each $\boldsymbol{\theta}$ and $\boldsymbol{\theta}' \in \Theta(\epsilon)$,

$$\mathbb{E}[\bar{\mathcal{G}}(\boldsymbol{\theta})] = 0 \quad \text{and} \quad \mathbb{E}[\bar{\mathcal{G}}(\boldsymbol{\theta}) \bar{\mathcal{G}}(\boldsymbol{\theta}')] = \text{avar}[n^{-1/2} \nabla_{\omega} \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0), n^{-1/2} \nabla_{\omega} \bar{L}_n(\boldsymbol{\theta}', \boldsymbol{\omega}_0)];$$

- (ii) $n^{-1} \nabla_{\omega}^2 \bar{L}_n(\cdot, \boldsymbol{\omega}_0) \xrightarrow{\mathbb{P}} \bar{\mathbf{A}}_0(\cdot)$, where for each $\boldsymbol{\theta} \in \Theta(\epsilon)$, $\bar{\mathbf{A}}_0(\boldsymbol{\theta}) := n^{-1} \mathbb{E}[\nabla_{\omega}^2 \bar{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)]$. □

Remark 2. (i) $\bar{\mathcal{G}}(\cdot)$ consists of $1 + r_2$ number of functions, and only its first element is a function of $\boldsymbol{\theta}$.

The others are independent of θ . Therefore, we may partition $\bar{\mathcal{G}}(\cdot)$ as follows:

$$\bar{\mathcal{G}}(\cdot) \equiv \begin{bmatrix} \bar{\mathcal{G}}_\pi(\cdot) \\ \bar{\mathcal{G}}_\xi \end{bmatrix}.$$

(ii) If $\mathcal{M}_o$ is correctly specified, the information matrix equality holds. That is, for each θ ,

$$\text{avar}[n^{-1/2}\nabla_\omega \bar{L}_n(\theta, \omega_0), n^{-1/2}\nabla_\omega \bar{L}_n(\theta, \omega_0)] = -\bar{\mathbf{A}}_0(\theta).$$

(iii) If we let $\theta = \theta_0$, $n^{-1/2}\nabla_\pi \ddot{L}_n(\theta_0, \omega_0) = 0$ and $n^{-1}\nabla_\pi^2 \ddot{L}_n(\theta_0, \omega_0) = 0$, so that the asymptotic distribution of $\{\nabla_\pi^2 \bar{L}_n(\theta_0, \omega_0)\}^{-1/2}\nabla_\pi \bar{L}_n(\theta_0, \omega_0)$ cannot be determined in the same way as for $\theta \in \Theta(\epsilon)$. In the next section, we examine the asymptotic distribution by removing the ϵ -restriction. $\square$

We now exploit Lemma 1 to obtain the limit distribution of the LR test under $\mathcal{H}_{02}$. We contain it in the following theorem:

Theorem 4. *Given Assumptions 1, 2, 6, 7, and 8, for each $\epsilon > 0$, $\mathcal{LR}_n''(\epsilon) \rightsquigarrow \sup_{\theta \in \Theta(\epsilon)} \max[0, \mathcal{Y}(\theta)]^2$ under $\mathcal{H}_{02}$, where for each θ ,*

$$\mathcal{Y}(\theta) := \{-\ddot{A}_{\pi\pi 0}(\theta)\}^{-1/2}\ddot{\mathcal{G}}_\pi(\theta), \quad \ddot{A}_{\pi\pi 0}(\theta) := \text{plim}_{n \rightarrow \infty} n^{-1}\nabla_\pi^2 \ddot{L}_n(\theta, \omega_0), \quad \text{and}$$

$$\ddot{\mathcal{G}}_\pi(\theta) := \bar{\mathcal{G}}_\pi(\theta) - \mathbb{E}[\nabla_\pi \nabla'_\xi \bar{L}_n(\theta, \omega_0)] \{\mathbb{E}[\nabla_\xi^2 \bar{L}_n(\theta, \omega_0)]\}^{-1} \bar{\mathcal{G}}_\xi. \quad \square$$

Therefore, the LR test weakly converges to a functional of a Gaussian process under $\mathcal{H}_{02}$.

Remark 3. (i) The null weak limit of the LR test under $\mathcal{H}_{02}$ is different from that of $\mathcal{LR}_n$ under $\mathcal{H}_0$. This aspect implies that the effect of model expansion is signaled by $\mathcal{LR}_n''(\epsilon)$ unlike $\mathcal{LR}_n'$. As the model expansion is effective, a different null limit distribution could be obtained from that of $\mathcal{LR}_n$.

(ii) The asymptotic distribution of the LR test is dependent upon the parameter space $\Theta(\epsilon)$. If Θ is replaced by a larger inclusive set, say Θ' , then it is apparent that $\sup_{\theta \in \Theta(\epsilon)} \max[0, \mathcal{Y}(\theta)]^2 \leq \sup_{\theta \in \Theta'(\epsilon)} \max[0, \mathcal{Y}(\theta)]^2$, so that the asymptotic test distribution varies. This implies that the asymptotic distributions have to be obtained separately for different parameter spaces. Having a larger parameter space corresponds to adding more alternatives to the original models, and it has a higher chance to include the DGP. Nevertheless, it does not mean that we can allow for an unbounded parameter space for θ . The asymptotic tightness condition is needed to derive the null limit distribution in Theorem 4, which holds when the parameter space is bounded. Without showing the asymptotic

tightness first, Theorem 4 may not hold.

- (iii) The covariance kernel form depends on the model, so that the null limit distribution of the LR test differs across models, meaning that the LR test is not distribution-free. There are a number of ways in the prior literature to obtain the approximate critical values as alternatives to obtain the exact asymptotic critical values. For example, if the covariance structure can be represented as an infinite sum of orthogonal functions, the null limit distribution of the LR test can be obtained straightforwardly by simulation (see [Cho and White, 2007](#); [Cho and Han, 2009](#); [Cho and White, 2010](#); [Cho, 2025, 2026](#), for example). As another example, conservative bounds of the tail distribution can be computed for the null limit distribution. [Davies \(1977\)](#) examines this problem in a general framework and provides the upper bounds of the asymptotic critical values using the up-crossing formula. [Piterbarg \(1996\)](#) provides another methodology to obtain the upper bound of the asymptotic critical values by comparing the covariance kernel with that of fractional Brownian motion. In Section 4 for Monte Carlo simulations, we illustrate a way to blend the simulation method and the bounding method to obtain the tail null limit distribution. □

3.3 Test under $\mathcal{H}_0$

In this section, we examine the null limit distribution of the LR test under $\mathcal{H}_0$.

The scores of the LR test obtained under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$ are not independent. This property may be important in obtaining the limit distribution of the LR test under $\mathcal{H}_0$, in which $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$ are nested. From the definition of the LR test:

$$\mathcal{LR}_n''' := \sup_{(\pi, \gamma) \in [0, 1] \times \Gamma} 2\{L_n(\pi, \gamma) - L_n^o(\tilde{\gamma}_n^o)\},$$

the weak limit of the LR test under $\mathcal{H}_0$ must be the maximum of the weak limits of the LR tests under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$. From this feature, it is necessary to identify the dependence structure between the weak limits in Theorems 3 and 4 to derive the null limit distribution of the LR test under $\mathcal{H}_0$.

Nevertheless, there exists a regular relationship between the weak limits of the LR tests under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$, and this can straightforwardly lead to the limit distribution of the LR test under $\mathcal{H}_0$. The following lemma provides their relationship:

Lemma 2. Given Assumptions 1, 2, 6, 7, and 8, if $\mathcal{M}_o$ is correctly specified,

$$\begin{aligned} & \lim_{\epsilon \downarrow 0} \sup_{\boldsymbol{\theta} \in \partial\boldsymbol{\Theta}(\epsilon)} \max \left[0, \{ -\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \}^{-1/2} \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \right]^2 \\ &= \nabla'_{\gamma} L_n^o(\gamma_0) \{ -\nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \mathbf{R}' [\mathbf{R}' \{ -\nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \mathbf{R}]^{-1} \mathbf{R} \{ -\nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \nabla_{\gamma} L_n^o(\gamma_0) \end{aligned} \quad (6)$$

under $\mathcal{H}_0$, where $\partial\boldsymbol{\Theta}(\epsilon) := \{ \boldsymbol{\theta} \in \boldsymbol{\Theta} : \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| = \epsilon \}$ and $\mathbf{R} := [\mathbf{I}_{r_1} \vdots \mathbf{0}_{r_1 \times r_2}]$. $\square$

Remark 4. (i) The left side of (6) maximizes the null approximation of the LR test under $\mathcal{H}_{02}$ over $\partial\boldsymbol{\Theta}(\epsilon)$ that converges to $\boldsymbol{\theta}_0$ as $\epsilon \rightarrow 0$. Meanwhile, the right side of (6) has the same asymptotic limit as the LR test obtained under $\mathcal{H}_{01}$. Therefore, an asymptotic equivalence is established between the null weak limits of the LR tests under $\mathcal{H}_{01}$ and $\mathcal{H}_{02}$. That is, as $\boldsymbol{\theta}$ converges to $\boldsymbol{\theta}_0$, $\{ -\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \}^{-1} \{ \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \}^2$ approaches $\{ -\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}_0, \boldsymbol{\omega}_0) \}^{-1} \{ \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}_0, \boldsymbol{\omega}_0) \}^2$ in a way the maximum is taken with respect to every direction around $\boldsymbol{\theta}_0$, so that it follows that

$$\lim_{\epsilon \downarrow 0} \mathcal{LR}_n''(\epsilon) \geq \mathcal{LR}_n'.$$

That is, if the ϵ -restriction is removed, the LR test obtained under $\mathcal{H}_{02}$ dominates the LR test under $\mathcal{H}_{01}$, so that the weak limit of the LR test under $\mathcal{H}_0$ can be obtained by the weak limit of the LR test without the ϵ -restriction under $\mathcal{H}_{02}$.

- (ii) When examining the left side of (6), careful analysis is required because $\lim_{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}_0} \{ -\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \}^{-1/2} \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)$ may not exist. For each $\boldsymbol{\theta}$, it can be equivalently represented as a function of the distance and direction between $\boldsymbol{\theta}$ and $\boldsymbol{\theta}_0$. As the distance tends to zero, the same limit may not be attained along every direction. Together with this aspect, $\{ -\nabla_{\pi}^2 \ddot{L}_n(\cdot, \boldsymbol{\omega}_0) \}^{-1/2} \nabla_{\pi} \ddot{L}_n(\cdot, \boldsymbol{\omega}_0)$ is certainly defined on an r_1 dimensional subspace $\boldsymbol{\Theta}$, so that approximating the numerator and denominator of $\{ -\nabla_{\pi}^2 \ddot{L}_n(\cdot, \boldsymbol{\omega}_0) \}^{-1/2} \nabla_{\pi} \ddot{L}_n(\cdot, \boldsymbol{\omega}_0)$ by Taylor's expansion involves r_1 number of arguments, and this may make the analysis complicated. However, we do not have to be concerned with all the directions endorsed by the r_1 number of arguments, because the 'sup' operator is employed with respect to the direction around $\boldsymbol{\theta}_0$ before ϵ tends to zero, which lets the limit of the ratio always exist as ϵ tends to zero. $\square$

Lemma 2 is crucial in obtaining the null limit distribution of the LR test under $\mathcal{H}_0$. As mentioned above, the weak limit of the LR test under $\mathcal{H}_0$ is dominated by the weak limit of the LR test without the ϵ -restriction under $\mathcal{H}_{02}$ by Lemma 2. We formally contain the implication of this statement in the following theorem:

Theorem 5. Given Assumptions 1, 2, 6, 7, and 8, if $\mathcal{M}_o$ is correctly specified, $\mathcal{LR}_n''' \rightsquigarrow \sup_{\theta \in \Theta} \max[0, \mathcal{Y}(\theta)]^2$ under $\mathcal{H}_0$. $\square$

Remark 5. (i) By Theorem 5, the limit distribution of the LR test under $\mathcal{H}_0$ is determined mainly by the Gaussian stochastic process obtained under $\mathcal{H}_{02}$. This aspect also implies that the effect of model expansion is signaled by $\mathcal{LR}_n'''$ as for $\mathcal{LR}_n''(\epsilon)$.
(ii) The null limit distribution of the LR test is similar to that obtained in the context of testing a mixture hypothesis, although a different hypothesis is tested from the mixture hypothesis. $\square$

4 Illustration and Monte Carlo Experiments

In this section, we illustrate the application of the LR test using a standard linear model. In particular, we focus on demonstrating how to obtain the asymptotic critical values of $\mathcal{LR}_n'''$ and comparing its size with those of $\mathcal{LR}_n$. We also compare their powers.

For Monte Carlo simulations, we consider the following model:

$$y_t = \alpha x_t + \beta z_t + u_t,$$

where $u_t \sim_{\text{i.i.d.}} \mathcal{N}(0, \tau)$, $\beta \in [L, U]$, and we suppose that either x_t or z_t is constant. If x_t is constant, then z_t is a random regressor, and vice versa. Given this, we further suppose that we test the following hypotheses:

$$\mathcal{H}_0 : \beta_* = 0 \quad \text{versus} \quad \mathcal{H}_1 : \beta_* \neq 0,$$

where $y_t | (x_t, z_t) \sim_{\text{i.i.d.}} \mathcal{N}(\alpha_* x_t, \tau_*)$ under $\mathcal{H}_0$, but the model is possibly misspecified under $\mathcal{H}_1$.

4.1 Critical Values

Under this setup, we specify the following mixture model to apply Theorem 5:

$$\pi \phi_t(\alpha, \beta, \tau) + (1 - \pi) \phi_t(\alpha, 0, \tau),$$

where $\phi_t(\alpha, \beta, \tau)$ is the conditional probability density function (PDF) of $y_t | (x_t, z_t)$, which is specified as $\mathcal{N}(\alpha x_t + \beta z_t, \tau)$. Given this, we can derive a conditional Gaussian process given $z_t = z$ using Lemma 1, having the following covariance structure under $\mathcal{H}_0$:

$$r(\beta, \beta' | z_t = z) := \frac{\exp\left(\frac{\beta\beta'}{\tau_*} z^2\right) - 1 - \frac{\beta\beta'}{\tau_*} z^2 - \frac{\beta^2\beta'^2}{2\tau_*^2} z^4}{\left\{\exp\left(\frac{\beta^2}{\tau_*} z^2\right) - 1 - \frac{\beta^2}{\tau_*} z^2 - \frac{\beta^4}{2\tau_*^2} z^4\right\}^{1/2} \left\{\exp\left(\frac{\beta'^2}{\tau_*} z^2\right) - 1 - \frac{\beta'^2}{\tau_*} z^2 - \frac{\beta'^4}{2\tau_*^2} z^4\right\}^{1/2}}.$$

If $z_t \equiv 1$, then $r(\beta, \beta'|z)$ is the unconditional covariance kernel $\mathcal{Y}(\cdot)$ given in Theorem 5. Otherwise, this should be interpreted as a conditional Gaussian process on $z_t = z$. The unconditional covariance structure can be different from this, and it depends upon the distribution of z_t . For example, if $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$, the unconditional covariance kernel is given as

$$\frac{\left(\frac{\tau_*}{\tau_* - 2\beta\beta'}\right)^{1/2} - 1 - \frac{\beta\beta'}{\tau_*} - \frac{3\beta^2\beta'^2}{2\tau_*^2}}{\left\{\left(\frac{\tau_*}{\tau_* - 2\beta^2}\right)^{1/2} - 1 - \frac{\beta^2}{\tau_*} - \frac{3\beta^4}{2\tau_*^2}\right\}^{1/2} \left\{\left(\frac{\tau_*}{\tau_* - 2\beta'^2}\right)^{1/2} - 1 - \frac{\beta'^2}{\tau_*} - \frac{3\beta'^4}{2\tau_*^2}\right\}^{1/2}}.$$

Requiring full distributional information on z_t can be restrictive, as many models are specified only for the conditional distribution $y_t|(x_t, z_t)$. We address this requirement by noting that the conditional covariance structure of $r(\beta, \beta'|z)$ is the same as that generated by $\mathcal{W}(\cdot|z_t = z)$, where for each β ,

$$\mathcal{W}(\beta|z_t = z) := \frac{\sum_{j=3}^{\infty} \frac{(\beta z)^j}{\sqrt{j!}\tau_*} w_j}{\left\{\exp\left(\frac{1}{\tau_*}\beta^2 z^2\right) - 1 - \frac{1}{\tau_*}\beta^2 z^2 - \frac{1}{2\tau_*^2}\beta^4 z^4\right\}^{1/2}}$$

and $w_j \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$. Therefore, it follows from this that

$$\mathcal{LR}_n''' \rightsquigarrow \sup_{\beta \in [L, U]} \max[0, \mathcal{Y}(\beta)]^2 = \sup_{\beta \in [\mathcal{Z}L, \mathcal{Z}U]} \max[0, \mathcal{W}(\beta|1)]^2 \quad (7)$$

in distribution, where $\mathcal{Z}$ is a random variable having the same marginal distribution as z_t and independent of $\mathcal{W}(\cdot)$.

- Remark 6.** (i) The stochastic feature of z_t results in the stochastic space for β , viz., $[\mathcal{Z}L, \mathcal{Z}U]$. Here, the conditional Gaussian process $\mathcal{W}(\cdot|1)$ is a complete orthogonal process, so that it is not difficult to simulate this process to obtain the test critical values. Therefore, the critical values of the LR test can be obtained by applying the simulation method, if the effect of the stochastic parameter space to the null limit distribution is appropriately clarified.
- (ii) The measurability of the right side follows by [Stinchcombe and White \(1992, theorem 2.17\)](#) or [Debreu \(1967, theorem 4.5\)](#). $\square$

We consider combining the simulation method with a procedure for obtaining conservative critical values of the LR test to account for the effect of the stochastic parameter space. [Piterbarg's \(1996\)](#) tail approximation of the null limit distribution can be applied to handle the stochastic parameter space. We first note the following theorem rephrased from theorem 7.1 of [Piterbarg \(1996\)](#):

Theorem 6. ([Piterbarg, 1996](#)) Suppose that a Gaussian stochastic process $\mathcal{X}(\cdot)$ has zero mean and covariance $r(\beta, \beta')$ such that for all $\varepsilon > 0$, there are α and $\delta > 0$, $|(1 - \varepsilon)(\beta - \beta')|^\alpha \leq 1 - r(\beta, \beta') \leq |(1 + \varepsilon)(\beta - \beta')|^\alpha$

if $|\beta - \beta'| < \delta$. Then if for $\beta \neq \beta'$, $r(\beta, \beta') < 1$, then

$$\mathbb{P} \left(\sup_{\beta \in [L, U]} \{\max[0, \mathcal{X}(\beta)]\}^2 > u^2 \right) = H_\alpha \lambda([L, U]) u^{2/\alpha} (1 - \Phi(u)) (1 + o(1)),$$

as $u \rightarrow \infty$, where $\lambda(\cdot)$ denotes the Lebesgue measure; $\Phi(\cdot)$ is the standard normal CDF; and $H_\alpha([0, \bar{U}]) := \mathbb{E}[\exp(\max_{\beta \in [0, \bar{U}]} \mathcal{B}^F(\beta))]$; $H_\alpha := \lim_{\bar{U} \rightarrow \infty} H_\alpha([0, \bar{U}])/\bar{U}$; and $\mathcal{B}^F(\cdot)$ is a fractional Gaussian stochastic process with mean $-|\beta|^\alpha$, and $\text{cov}(\mathcal{B}^F(\beta), \mathcal{B}^F(\beta')) = |\beta|^\alpha + |\beta'|^\alpha - |\beta - \beta'|^\alpha$. $\square$

We do not prove Theorem 6 because it follows from Piterbarg (1996). By Theorem 6, it is now apparent that as $u \rightarrow \infty$,

$$\mathbb{P} \left(\sup_{\beta \in [L, U]} \{\max[0, \mathcal{W}(\beta|\mathcal{Z})]\}^2 > u^2 \right) = H_\alpha \mathbb{E}[\lambda([\mathcal{Z}L, \mathcal{Z}U])] u^{2/\alpha} (1 - \Phi(u)) (1 + o(1))$$

from the fact that $r(\beta, \beta'|z)$ is twice continuously differentiable at $\beta = \beta' \neq 0$ a.s. This implies that the tail null limit distribution of the LR test can be well approximated when the stochastic parameter space in (7) is replaced by its expectation, viz.,

$$P_E := [\min\{\mathbb{E}[z_t]L, \mathbb{E}[z_t]U\}, \max\{\mathbb{E}[z_t]L, \mathbb{E}[z_t]U\}].$$

We now combine this with the simulation method to obtain the tail null limit distribution of the LR test. That is, we simulate $\sup_{\beta \in P_E} \max[0, \mathcal{W}(\beta|1)]^2$ to obtain the asymptotic critical values. If P_E is unknown, we can consistently estimate it.

4.2 Size and Power of LR Test

In this section we examine the size and power properties of the LR test by simulation.

For simulation, we consider two DGPs for z_t : $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} U[-0.5, 0.5]$. These choices allow us to investigate how sensitive the asymptotic null distributions are to the stochastic parameter space. We suppose that the initial parameter space is given by $L = -2$ and $U = 2$. Then, P_E is given as $[-2\sqrt{2/\pi}, 2\sqrt{2/\pi}]$ and $[-0.5, 0.5]$ for the first and the second DGPs, respectively.

If the parameter space P_E is unknown, we need to estimate it. For this, we first let $\hat{\nu}_n := n^{-1} \sum_{t=1}^n |z_t|$, and estimate P_E by $\hat{P}_{E,n} := [\hat{\nu}_n L, \hat{\nu}_n U]$, so that the tail null limit distribution of the LR test can be generated by simulating $\sup_{\beta \in \hat{P}_{E,n}} \max[0, \mathcal{W}(\beta|1)]^2$. We report the asymptotic critical values in Table 1 obtained in this manner. We compute the asymptotic critical values for various sample sizes, 50, 100, 500, and 1000 to compare them with the critical values obtained when P_E is known. The last column of Table 1 denotes

the case when P_E is used to simulate the critical values. They are obtained by repeating 10000 independent experiments. From this, we observe that the critical values obtained by $\hat{P}_E$ are quite precise, although they are more or less different from when $\mathbb{E}[|z_t|]$ is known. If the level of significance is small, the difference gets larger. For example, if the level of significance is 1%, the difference is more negligible when the sample size is large (say $n = 1000$) than the other cases with smaller sample sizes (say $n = 50$). However, as the level of significance increases, their difference becomes more negligible irrespective of the sample size.

We next examine the size of the LR test $\mathcal{LR}_n'''$ using the critical values described above. Table 2 reports the empirical sizes of the LR test, showing that the nominal levels are well controlled even though the critical values are obtained from the tail null limit distribution.

We compare the empirical distribution of the LR test $\mathcal{LR}_n'''$ with the empirical distribution obtained by the tail null limit distribution. The primary goal of this comparison is to examine how well the tail null limit distribution approximates the finite-sample null distribution of the LR test statistic $\mathcal{LR}_n'''$. This is important in choosing an appropriate level of test. If the tail approximation is effective only for extremely small levels, then a small significance level has to be selected. Otherwise, the critical values obtained by the tail null limit distribution have to be regarded as conservative critical values.

Figures 1 and 2 show the empirical distribution of the LR test $\mathcal{LR}_n'''$ and the tail null limit distribution. Specifically, Figures 1 and 2 compare the simulation method based empirical distributions with the empirical distributions of the LR test obtained under the assumption that $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$, respectively. For brevity, we provide the distributions only for the case when n is 50 and 500. We could obtain similar results for other sample sizes. Under our model assumptions, the empirical distributions of the LR test are well recovered by the tail null limit distribution. This implies that the level of test can be chosen without being concerned with the tail approximation condition for the current model case.

Finally, we examine the empirical power of the LR test $\mathcal{LR}_n'''$ using the same critical values and compare it with that of the standard LR test $\mathcal{LR}_n$. For this goal, we assume the following DGP condition:

$$(y_t|z_t) \sim_{\text{i.i.d.}} \pi_* \mathcal{N}(\beta_* z_t, 1) + (1 - \pi_*) \mathcal{N}(0, 1),$$

where $\beta_* \in \{0.3, 0.6, 0.9, 1.2, 1.5\}$, $\pi_* \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, and $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ or $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$.

There are a couple of remarks regarding the alternative DGPs. First, given the DGP condition, the model assuming a single component is misspecified, so that the extremum estimator is not ML but quasi-maximum likelihood (QML) estimator. The same applies for the LR test. We should call it quasi-LR (QLR) test to distinguish it from $\mathcal{LR}_n'''$. Second, if the single-component based model is correctly specified, $\mathcal{LR}_n$ is locally

most powerful test, so that $\mathcal{LR}_n'''$ cannot be more powerful than $\mathcal{LR}_n$. This aspect has to be expected from Monte Carlo simulations. The goal to form the mixture model is not to have uniformly most powerful test but to complement the standard LR test which may have poor power by model misspecification. As π_* tends to one, the given DGP converges to the single-component based DGP, so that $\mathcal{LR}_n$ should be locally the most powerful test, meaning that $\mathcal{LR}_n$ is more powerful than $\mathcal{LR}_n'''$. This aspect implies that the power behavior of $\mathcal{LR}_n'''$ is expected to critically depend upon the level of π_* .

Simulation results are contained in Table 3. The empirical powers are obtained by repeating 2500 independent experiments with $n = 100$. Table 3 reports the case in which $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$. The simulation results are nuanced, but this nuance is consistent with our expectation. When $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$, $\mathcal{LR}_n'''$ is more powerful than $\mathcal{LR}_n$ for π_* close to zero, and the $\mathcal{LR}_n$ is more powerful for π_* close one. The border line between these two different power behaviors appears to be around between 0.5 and 0.7. On the other hand, when $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$, $\mathcal{LR}_n'''$ is superior to or exhibits power comparable to that of $\mathcal{LR}_n$. This holds even when π_* is close to unity. Although their difference is not so large enough, this aspect implies that $\mathcal{LR}_n'''$ can be effective in detecting the alternatives.

5 Conclusion

In this study, we examine the testing of standard hypotheses by forming a mixture of null and alternative models. The main motivation for this model expansion stems from the concern that the null hypothesis may be rejected simply because the original model is misspecified. We address this concern by expanding the model scope, focusing on a mixture model formed by combining the null and alternative models such that the original model is nested within it.

The LR test based on the mixture framework requires non-standard analysis under the null hypothesis because the model fails to satisfy the standard regularity conditions. Specifically, the model suffers from twofold identification and boundary parameter problems under the null. We overcome these challenges and demonstrate that the null limit distribution of the LR test is represented as a functional of a Gaussian stochastic process. We illustrate the implementation of the LR test using a standard linear model and demonstrate its usefulness through Monte Carlo simulations. Furthermore, we show via simulations that the power of the mixture-based LR test can surpass that of the standard LR test when the model is misspecified in the direction of the mixture.

6 Appendix

Proof of Theorem 2: We can apply the weak ULLN to prove this. By the differentiability condition in Assumption 2, Lipschitz condition holds, so that

$$|L_n(\pi, \gamma) - L_n(\pi', \gamma')| \leq D_n \|(\pi, \gamma) - (\pi', \gamma')\|,$$

where $D_n := \sup_{(\pi, \gamma)} \|\nabla_{(\pi, \gamma)} L_n(\pi, \gamma)\|_\infty$, which is dominated by $\sum_{t=1}^n M_t$ by Assumption 7. Therefore, for any $\varepsilon > 0$,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(n^{-1} |L_n(\pi, \gamma) - L_n(\pi', \gamma')| > \varepsilon) \leq \limsup_{n \rightarrow \infty} \mathbb{P}(n^{-1} D_n \|(\pi, \gamma) - (\pi', \gamma')\| > \varepsilon),$$

and we can apply the ergodic theorem to $n^{-1} D_n$, so that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(n^{-1} |L_n(\pi, \gamma) - L_n(\pi', \gamma')| > \varepsilon) < \varepsilon,$$

whenever $\|(\pi, \gamma) - (\pi', \gamma')\| < \varepsilon/D_*$, where $D_* := \text{plim } n^{-1} D_n$. Therefore, $\{n^{-1} L_n(\cdot, \cdot)\}$ is stochastically equicontinuous, which implies that $n^{-1} L_n(\cdot, \cdot)$ uniformly converges on $[0, 1] \times \mathbf{\Gamma}$. The pointwise ergodic theorem implies that the limit of $n^{-1} L_n(\cdot, \cdot)$ is $\mathbb{E}[n^{-1} L_n(\cdot, \cdot)]$. This completes the proof. ■

Proof of Theorem 3: The proof is straightforward by noting that

$$2\{L_n^o(\tilde{\gamma}_n^n) - L_n^o(\gamma_0)\} = \{\nabla'_\xi L_n^o(\gamma_0)\} \{-\nabla_\xi^2 L_n^o(\gamma_0)\}^{-1} \{\nabla_\xi L_n^o(\gamma_0)\} + o_{\mathbb{P}}(1)$$

by a second-order Taylor's expansion, so that

$$\begin{aligned} \mathcal{LR}'_n &= \nabla'_\gamma L_n^o(\gamma_0) \{-\nabla_\gamma^2 L_n^o(\gamma_0)\}^{-1} \nabla_\gamma L_n^o(\gamma_0) - \nabla'_\xi L_n^o(\gamma_0) \{-\nabla_\xi^2 L_n^o(\xi_0)\}^{-1} \nabla_\xi L_n^o(\gamma_0) + o_{\mathbb{P}}(1) \\ &= 2\{L_n^o(\hat{\gamma}_n^n) - L_n^o(\gamma_0)\} - 2\{L_n^o(\tilde{\gamma}_n^n) - L_n^o(\gamma_0)\} + o_{\mathbb{P}}(1) = \mathcal{LR}_n + o_{\mathbb{P}}(1) \rightsquigarrow \mathbf{z}' \mathbf{z} \end{aligned}$$

by Theorem 1. This completes the proof. ■

Proof of Lemma 1: (i) To show, we note that only the first element of $n^{-1/2} \nabla_\omega \tilde{L}_n(\cdot, \omega_0)$ is a function of θ . The asymptotic tightness follows from this easily. First, the finite dimensional central limit theorem holds given our regularity conditions and theorem 1 of Doukhan and Rio (1995). That is, for any $\theta^{(i)} \neq \theta^{(j)}$ and

$\boldsymbol{\theta}^{(i)} \in \boldsymbol{\Theta}(\epsilon)$ with $i, j = 1, \dots, m \in \mathbb{N}$,

$$\{n^{-1/2}\nabla_{\pi}\tilde{L}_n(\boldsymbol{\theta}^{(1)}, \boldsymbol{\omega}_0), \dots, n^{-1/2}\nabla_{\pi}\tilde{L}_n(\boldsymbol{\theta}^{(m-1)}, \boldsymbol{\omega}_0), n^{-1/2}\nabla_{\pi}\tilde{L}_n(\boldsymbol{\theta}^{(m)}, \boldsymbol{\omega}_0)\} \\ \rightsquigarrow \{\tilde{\mathcal{G}}_{\pi}(\boldsymbol{\theta}^{(1)}), \dots, \tilde{\mathcal{G}}_{\pi}(\boldsymbol{\theta}^{(m-1)}), \tilde{\mathcal{G}}_{\pi}(\boldsymbol{\theta}^{(m)})\}.$$

Second, for this step, we mainly rely on Hansen's (2006, proof of lemma 1) approach. We let $H(u)$ be the entropy with bracketing in $L^{2+\varsigma}$ norm for some $\varsigma > 0$. Given this, if $\int_0^1 \sqrt{H(u)} du < \infty$, the weak convergence follows from theorem 1 of Doukhan and Rio (1995). Given our regularity condition, for any $\eta > 0$, if $\varsigma \leq 2\delta$,

$$E \left\{ \sup_{\|\boldsymbol{\theta} - \boldsymbol{\theta}'\| < \eta} \|\nabla_{\pi}\ell_t(\boldsymbol{\theta}, \boldsymbol{\omega}_0) - \nabla_{\pi}\ell_t(\boldsymbol{\theta}', \boldsymbol{\omega}_0)\|^{2+\varsigma} \right\} \leq \Delta\eta.$$

by the differentiability of $\nabla_{\pi}\ell_t(\cdot, \boldsymbol{\omega}_0)$ and the existence of its bound M_t in Assumption 7. Theorem 6 of Andrews (1994) therefore implies that $n^{-1/2}\tilde{L}_n(\cdot, \boldsymbol{\omega}_0)$ weakly converges to a Gaussian process whose covariance structure is given as in the claim, as desired.

(ii) By Assumption 7, $\sup_{(\boldsymbol{\theta}, \boldsymbol{\omega})} \|\nabla_{\omega}^2 \tilde{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega})\|_{\infty} \leq \sum M_t$, where $n^{-1} \sum M_t$ obeys the ergodic theorem. Ranga Rao (1962) completes the proof. ■

Proof of Theorem 4: Given Lemma 1, we can apply the continuous mapping theorem to (5). Then

$$\begin{aligned} \mathcal{LR}_n''(\epsilon) &= \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}(\epsilon)} 2\{\bar{L}_n(\boldsymbol{\theta}, \hat{\boldsymbol{\omega}}_n(\boldsymbol{\theta})) - \bar{L}_n(\boldsymbol{\theta}, \tilde{\boldsymbol{\omega}}_n^o)\} \\ &= \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}(\epsilon)} \max[0, \{-\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)\}^{-1/2} \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)]^2 + o_{\mathbb{P}}(1) \\ &\rightsquigarrow \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}(\epsilon)} \max[0, \{-\ddot{A}_{\pi\pi*}(\boldsymbol{\theta})\}^{-1/2} \ddot{\mathcal{G}}_{\pi}(\boldsymbol{\theta})]^2 = \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}(\epsilon)} \max[0, \mathcal{Y}(\boldsymbol{\theta})]^2, \end{aligned}$$

as desired. This completes the proof. ■

Proof of Lemma 2: Note that $\lim_{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}_0} \nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) = 0$ and $\lim_{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}_0} \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) = 0$. We therefore, exploit l'Hopital's rule to obtain the limit. Then

$$\frac{\max[0, \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)]^2}{\{-\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)\}} = \frac{\max[0, (\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \mathbf{R} \{\nabla_{\gamma}^2 L_n^o(\gamma_0)\}^{-1} \nabla_{\gamma} L_n^o(\gamma_0)]^2}{(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \mathbf{R}' \{-\nabla_{\gamma}^2 L_n^o(\gamma_0)\}^{-1} \mathbf{R}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)} + o_{\mathbb{P}}(\epsilon^2),$$

Given this,

$$(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \{\mathbf{R}' \{-\nabla_{\gamma}^2 L_n^o(\gamma_0)\}^{-1} \mathbf{R}\}^{1/2} [(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \mathbf{R}' \{-\nabla_{\gamma}^2 L_n^o(\gamma_0)\}^{-1} \mathbf{R}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)]^{-1/2}$$

is a unit vector for any $\boldsymbol{\theta}$, so that $\max[0, \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)]^2 / \{-\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)\} = \lambda_{\max}[\mathbf{U}_n \mathbf{U}_n'] + o_{\mathbb{P}}(\epsilon^2)$, if

$(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \mathbf{R} \{ \nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \nabla_{\gamma} L_n^o(\gamma_0) \geq 0$, where

$$\mathbf{U}_n := \{ \mathbf{R}' \{ -\nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \mathbf{R} \}^{-1/2} \{ \mathbf{R} \{ \nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \nabla_{\gamma} L_n^o(\gamma_0) \}.$$

Otherwise, it is $o_{\mathbb{P}}(\epsilon^2)$. Further, $\text{rank}(\mathbf{U}_n \mathbf{U}_n') = 1$, and so $\text{tr}(\mathbf{U}_n \mathbf{U}_n') = \mathbf{U}_n' \mathbf{U}_n = \lambda_{\max}[\mathbf{U}_n \mathbf{U}_n']$. Therefore,

$$\sup_{\boldsymbol{\theta} \in \partial \boldsymbol{\Theta}(\epsilon)} \max[0, \nabla_{\pi} \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0)]^2 / \{ -\nabla_{\pi}^2 \ddot{L}_n(\boldsymbol{\theta}, \boldsymbol{\omega}_0) \} = \mathbf{U}_n' \mathbf{U}_n + o_{\mathbb{P}}(\epsilon^2)$$

because $(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$ can be always chosen in a way $(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \mathbf{R} \{ \nabla_{\gamma}^2 L_n^o(\gamma_0) \}^{-1} \nabla_{\gamma} L_n^o(\gamma_0) \geq 0$. This completes the proof. $\blacksquare$

Proof of Theorem 5: From the definition of the LR test,

$$\begin{aligned} \mathcal{LR}_n''' &= \lim_{\epsilon \rightarrow 0} \max[\mathcal{LR}_n'', \mathcal{LR}_n''(\epsilon)] = \lim_{\epsilon \rightarrow 0} \mathcal{LR}_n''(\epsilon) + o_{\mathbb{P}}(1) \\ &\rightsquigarrow \lim_{\epsilon \rightarrow 0} \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}(\epsilon)} \max[0, \mathcal{Y}(\boldsymbol{\theta})]^2 = \sup_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} \max[0, \mathcal{Y}(\boldsymbol{\theta})]^2, \end{aligned}$$

where the second equality follows from Lemma 2, and the weak convergence follows from Theorem 4. The final equality holds by the separability of $\mathcal{Y}(\cdot)$. This completes the proof. $\blacksquare$

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Nominal Level \ n	50	100	500	1,000	∞
1.00 %	7.9342	8.2230	7.9793	7.7948	7.7613
2.50 %	6.1945	6.2026	6.1830	6.2187	6.1573
5.00 %	4.8272	4.9859	4.9710	4.8882	4.7457
7.50 %	4.1595	4.1918	4.2621	4.1300	4.0955
10.0 %	3.6048	3.6882	3.7321	3.6424	3.5944
12.5 %	3.1919	3.3281	3.3114	3.2872	3.1792
15.0 %	2.8726	3.0076	2.9931	2.9509	2.9053
1.00 %	6.8793	7.1947	7.0726	7.0413	7.0464
2.50 %	5.3965	5.3405	5.4058	5.4515	5.2720
5.00 %	4.1453	4.1954	4.2921	4.1815	4.0734
7.50 %	3.4280	3.5145	3.5893	3.4857	3.4256
10.0 %	2.9452	3.0184	3.0987	3.0318	2.9718
12.5 %	2.5583	2.6659	2.6552	2.6462	2.6388
15.0 %	2.2754	2.3525	2.3749	2.3577	2.3438

Table 1: CRITICAL VALUES. The above critical values of the LR test are obtained by assuming the following: The first panel assumes DGP: $(y_t|z_t) \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and model: $(y_t|z_t) \sim \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$ and $\beta \in [-2, 2]$. The second panel assumes DGP: $(y_t|z_t) \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$ and model: $(y_t|z_t) \sim \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$ and $\beta \in [-2, 2]$. Total number of iterations is 10000.

Nominal Level \ n	50	100	200	300	400	500
1.00	1.52	1.36	1.09	1.29	1.35	1.24
5.00	6.74	6.08	6.09	6.18	6.51	6.47
10.0	11.86	10.91	11.22	11.57	11.55	11.59
1.00	1.11	0.96	1.01	0.96	1.12	1.11
5.00	5.96	5.91	5.70	5.74	5.92	5.86
10.0	11.17	10.63	10.52	10.77	10.89	10.76

Table 2: EMPIRICAL REJECTION RATES OF THE LR TEST UNDER THE NULL (IN PERCENT). The empirical rejection rates of the LR test are obtained by assuming the following: The first panel assumes DGP: $(y_t|z_t) \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and model: $(y_t|z_t) \sim \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$ and $\beta \in [-2, 2]$. The second panel assumes DGP: $(y_t|z_t) \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$ and model: $(y_t|z_t) \sim \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$ and $\beta \in [-2, 2]$. Total number of iterations is 10000.

π_*	LR Test \ β_*	0.3	0.6	0.9	1.2	1.5
0.1	$\mathcal{LR}_n$	7.00	9.72	15.80	23.52	29.68
	$\mathcal{LR}_n'''$	7.76	14.56	22.12	40.16	53.76
0.3	$\mathcal{LR}_n$	15.04	40.56	68.36	83.88	92.64
	$\mathcal{LR}_n'''$	16.72	44.32	74.40	90.88	97.40
0.5	$\mathcal{LR}_n$	32.40	79.60	96.60	99.40	99.92
	$\mathcal{LR}_n'''$	31.88	79.92	98.08	99.84	100.0
0.7	$\mathcal{LR}_n$	55.16	96.64	100.0	100.0	100.0
	$\mathcal{LR}_n'''$	52.68	96.76	100.0	100.0	100.0
0.9	$\mathcal{LR}_n$	75.40	99.80	100.0	100.0	100.0
	$\mathcal{LR}_n'''$	70.52	99.76	100.0	100.0	100.0
0.1	$\mathcal{LR}_n$	5.24	5.72	5.56	6.60	7.48
	$\mathcal{LR}_n'''$	5.84	6.44	6.08	7.04	8.48
0.3	$\mathcal{LR}_n$	6.04	8.92	12.12	17.40	25.96
	$\mathcal{LR}_n'''$	6.40	9.32	12.24	18.64	27.68
0.5	$\mathcal{LR}_n$	6.60	15.36	25.28	39.92	56.12
	$\mathcal{LR}_n'''$	7.08	15.84	26.32	40.72	57.12
0.7	$\mathcal{LR}_n$	9.36	22.84	44.48	65.56	83.12
	$\mathcal{LR}_n'''$	10.24	24.20	45.36	66.16	83.60
0.9	$\mathcal{LR}_n$	13.00	33.28	63.60	86.40	96.92
	$\mathcal{LR}_n'''$	13.52	34.52	63.80	86.28	96.92

Table 3: EMPIRICAL REJECTION RATES OF THE LR TEST UNDER THE ALTERNATIVE (IN PERCENT, 5% NOMINAL LEVEL). The empirical rejection rates of the LR tests are obtained by assuming the following: The first panel assumes DGP: $(y_t|z_t) = \pi_*\mathcal{N}(\beta_*z_t, 1) + (1 - \pi_*)\mathcal{N}(0, 1)$, $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and model: $(y_t|z_t) \sim_{\text{i.i.d.}} \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$, $\beta \in [-2, 2]$. The second panel assumes DGP: $(y_t|z_t) = \pi_*\mathcal{N}(\beta_*z_t, 1) + (1 - \pi_*)\mathcal{N}(0, 1)$, $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$ and model: $(y_t|z_t) \sim_{\text{i.i.d.}} \pi \cdot \mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi) \cdot \mathcal{N}(\alpha, \tau)$, $\beta \in [-2, 2]$. Total number of iterations is 2500.

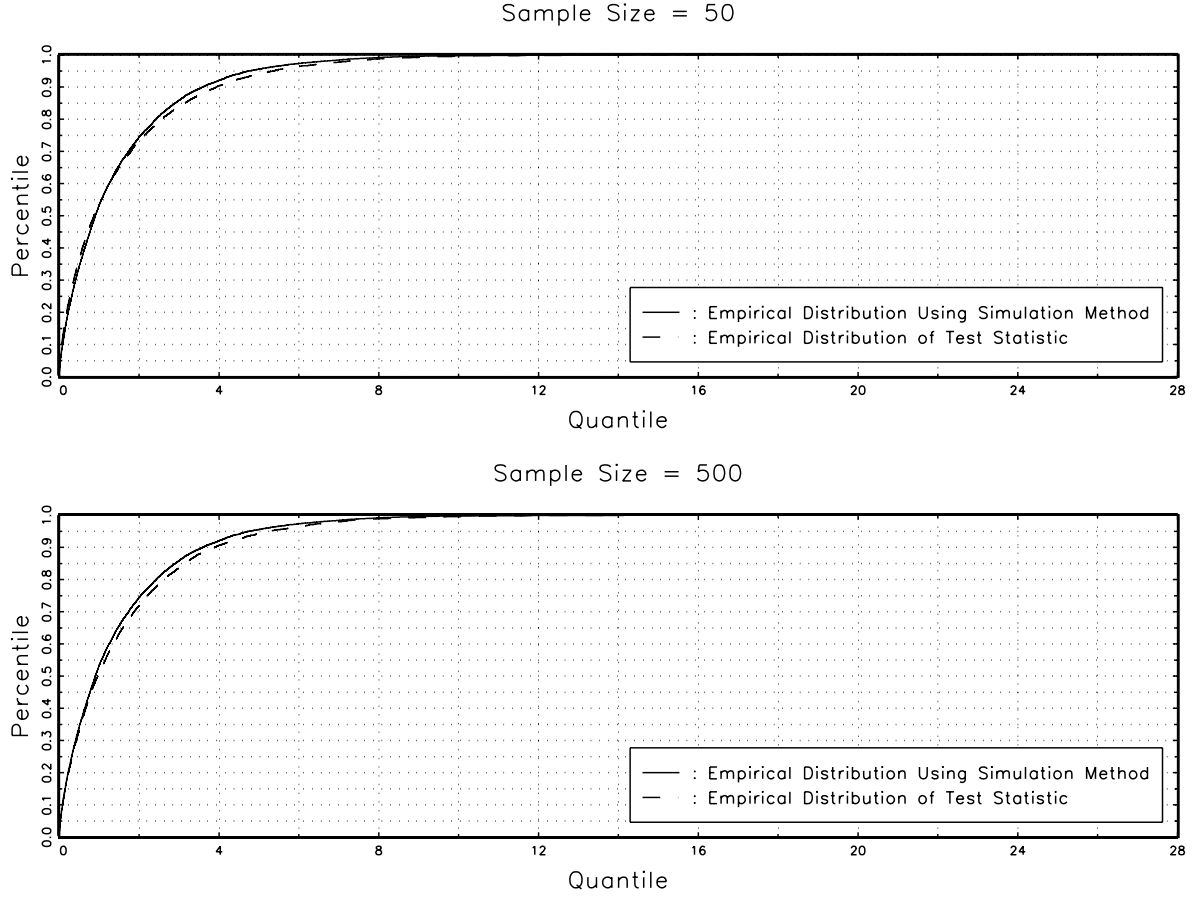


Figure 1: EMPIRICAL DISTRIBUTION AND APPROXIMATED NULL DISTRIBUTION. This figure draws the empirical Distribution and approximated null Distribution of the LR test by assuming the following: The first panel assumes DGP: $(y_t|z_t) \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and model: $(y_t|z_t) \sim \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$ and $\beta \in [-2, 2]$.

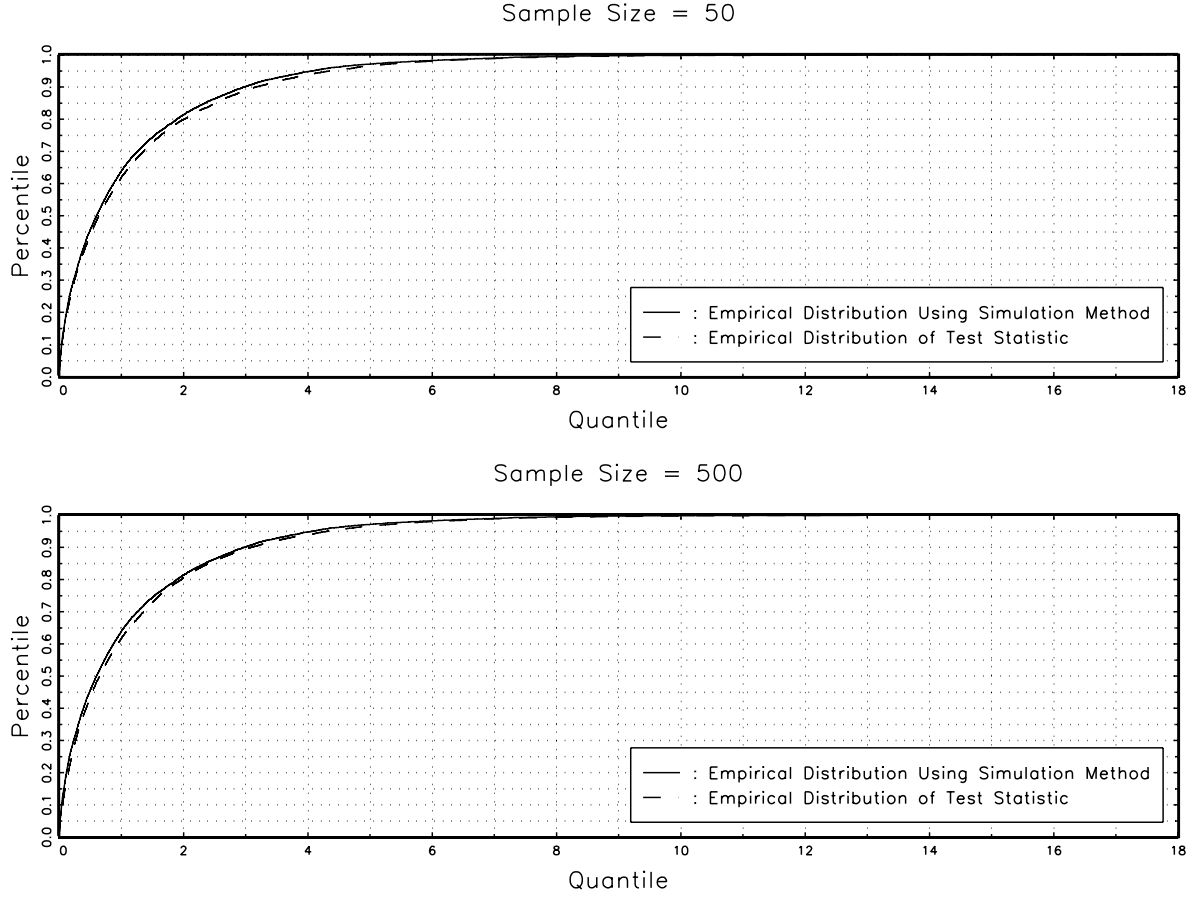


Figure 2: EMPIRICAL DISTRIBUTION AND APPROXIMATED NULL DISTRIBUTION. This figure draws the empirical Distribution and approximated null Distribution of the LR test by assuming the following: The first panel assumes DGP: $(y_t|z_t) \sim_{\text{i.i.d.}} \mathcal{N}(0, 1)$ and $z_t \sim_{\text{i.i.d.}} \mathcal{U}(-0.5, 0.5)$ and model: $(y_t|z_t) \sim \pi\mathcal{N}(\alpha + \beta z_t, \tau) + (1 - \pi)\mathcal{N}(\alpha, \tau)$ and $\beta \in [-2, 2]$.